

# Triple-drug Fixed-dose Combination of Dapagliflozin + Glimepiride + Metformin Extended Release: Rationale for Combination

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Neeraj Markandeywar,<sup>1</sup> Mandodari Rajurkar,<sup>1</sup> Shruti Dharmadhikari,<sup>1</sup> Sapan Behra,<sup>2</sup> Piyush Patel,<sup>2</sup> Dipak Patil,<sup>1</sup> Lalit Lakhwani,<sup>2</sup> Pravin Ghadge,<sup>1</sup> Suyog Mehta,<sup>1</sup> Sadhna Joglekar<sup>2</sup>

\*Presenting author 1: Sun Pharma Laboratories Limited, Mumbai, India 2: Ex-employees, Sun Pharma, Mumbai, India

## Epidemiology of Diabetes in India

- Globally, India ranks 2<sup>nd</sup> in the total number of diabetes cases<sup>1</sup>
- Diabetes prevalence was approximately 101 million in 2021<sup>2</sup>

## Unmet Needs in Management of Type 2 Diabetes Mellitus

Chronic high glucose levels can cause glucose toxicity in several organs leading to complications such as chronic kidney disease, heart disease, nerve damage.<sup>3,4</sup>

With diabetes, there is a progressive decline in beta-cell function. Therefore, many patients will eventually require multiple agents.<sup>5</sup>

Dual therapy (Metformin + SU/SGLT2i) with mean baseline HbA1c of 8%–8.9% has shown < 22% are able to achieve target HbA1c < 7%

In addition high pill burden and complex treatment regimens decrease adherence<sup>6</sup>

## Need for Fixed Dose Combinations (FDC)

Diabetes has multifactorial pathophysiologies.<sup>7</sup>  
Therefore, components of FDC should:

Act by complimentary mechanisms<sup>8</sup>

Should be well tolerated as FDC<sup>8</sup>

Triple-drug FDCs effectively achieve glycemic targets<sup>7</sup>

## Guideline Recommendations for Inadequately Controlled Diabetes with Metformin

## 2023 ADA Recommendations<sup>9</sup>

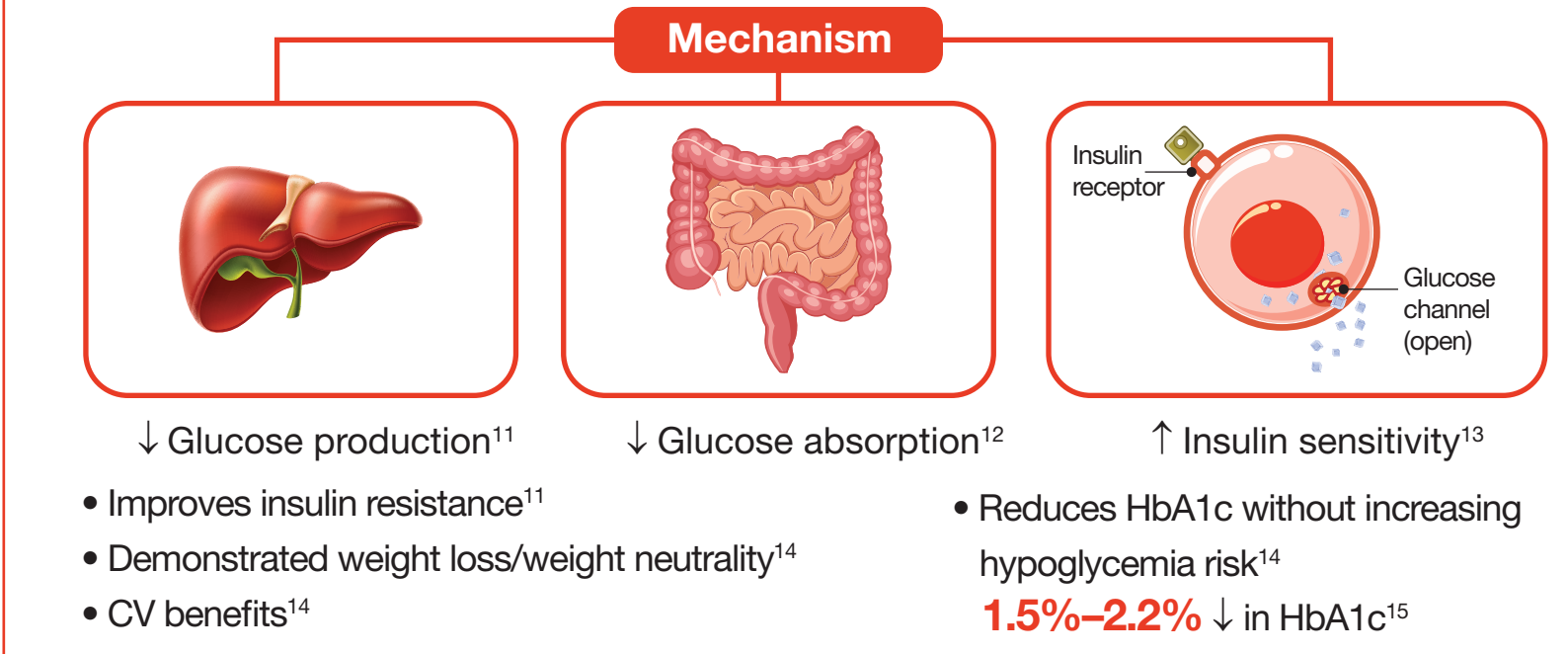
Pharmacologic approaches that provide the efficacy to achieve treatment goals should be considered, such as Metformin or other agents, including combination therapy, that provide adequate efficacy to achieve and maintain treatment goals

## 2020 RSSDI-ESI Clinical Practice Recommendations<sup>10</sup>

If glucose targets are not achieved with two agents: start **third oral agent**-AGI, DPP-4 inhibitor, SGLT2 inhibitor, or TZDs (depending on second-line agent used) or start insulin or GLP-1 agonists

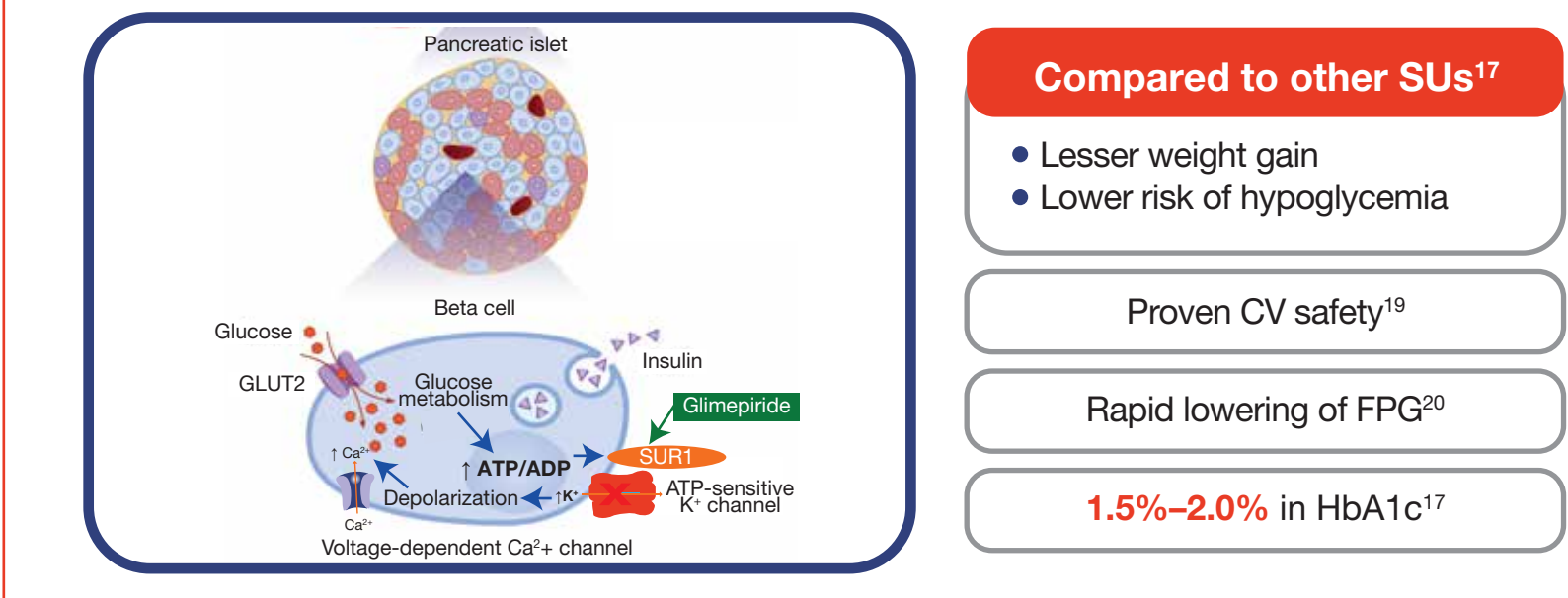
## Drug Profile – Metformin, Glimepiride, and Dapagliflozin

## Metformin – First-line Treatment for > 50 Years<sup>11</sup>

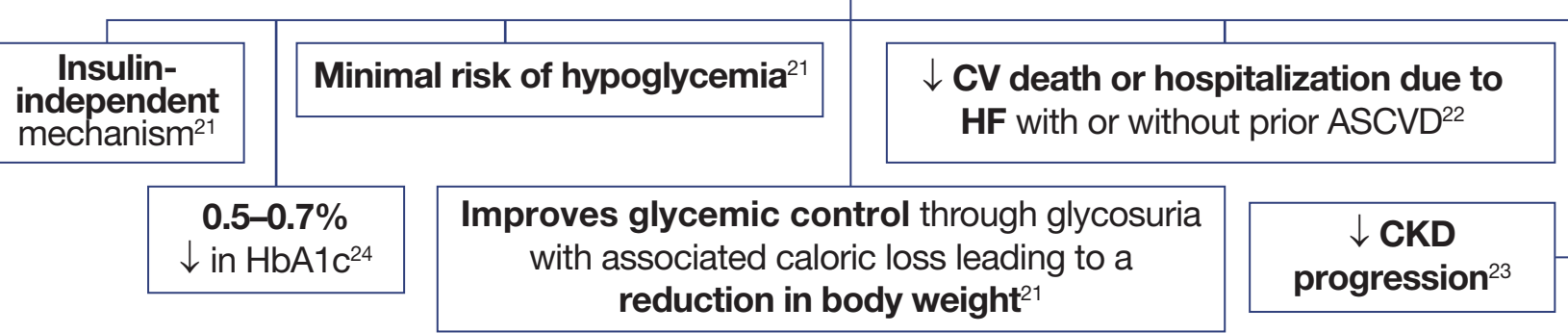


## Glimepiride – A Modern SU<sup>16</sup>

Increases insulin secretion through stimulation of pancreatic beta cells<sup>17,18</sup>



## Dapagliflozin – A Highly Selective SGLT2i<sup>21</sup>



## Advantages of triple-drug FDC

- Allows once-daily dosing<sup>25</sup>
- Drugs with complimentary mechanism of action<sup>25</sup>
- Enhance weight loss in overweight/obese patients<sup>7</sup>
- Cardiovascular benefits<sup>7</sup>

## Dapagliflozin + Glimepiride + Metformin FDC

- ↑ insulin sensitivity<sup>13</sup>
- ↓ insulin resistance<sup>11</sup>
- ↓ body weight<sup>21</sup>

## Evidence for Dapagliflozin + Glimepiride + Metformin (Data from 24-week Randomized, Double-blind, Placebo-controlled, Parallel-group, International, Multicenter Studies)

| Phase 3 study <sup>26</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Phase 3b study <sup>27</sup>                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li><b>Dapagliflozin vs Placebo</b><br/><b>HbA1c reduction from</b> baseline (<math>p &lt; 0.0001</math>):<ul style="list-style-type: none"><li>Dapagliflozin (2 mg) = <b>-0.58%</b></li><li>Dapagliflozin (5 mg) = <b>-0.63%</b></li><li>Dapagliflozin (10 mg) = <b>-0.82%</b></li></ul></li><li>Sustained <b>reduction in body weight</b> from baseline:<ul style="list-style-type: none"><li>Dapagliflozin (2 mg) = -1.18 kg (<math>p &lt; 0.1410</math>)</li><li>Dapagliflozin (5 mg) = -1.56 kg (<math>p &lt; 0.0091</math>)</li><li>Dapagliflozin (10 mg) = -2.26 kg (<math>p &lt; 0.0001</math>)</li></ul></li></ul> | <ul style="list-style-type: none"><li><b>Dapagliflozin (10 mg/day) vs Placebo</b><ul style="list-style-type: none"><li><b>Reduction in HbA1c</b> from baseline (<b>-0.86%</b> vs -0.17%, <math>p &lt; 0.0001</math>)</li><li><b>Improvement in FPG</b> from baseline (-34.2 mg/dl vs -0.8 mg/dl, <math>p &lt; 0.0001</math>)</li><li><b>Reduction in body weight</b> from baseline (-2.7 kg vs -0.6 kg, <math>p &lt; 0.0001</math>)</li></ul></li></ul> |
| Both the studies showed that Dapagliflozin when used in a combination therapy was effective, well-tolerated, and reduced body weight. <sup>26,27</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

## Conclusion

**Triple-drug FDC (Dapagliflozin + Glimepiride + Metformin) will provide better safety and efficacy due to:**

- Complimentary mechanisms of action
- Potential for reducing complications
- Less risk of hypoglycemia
- Additional benefit of weight loss
- Improved compliance

**Abbreviations:** ADA: American Diabetes Association; ADP: Adenosine di phosphate; AGI: Alpha-glucosidase inhibitors; ASCVD: Atherosclerotic cardiovascular disease; ATP: Adenosine tri phosphate; CKD: Chronic kidney disease; CV: Cardiovascular; DPP4i: Dipeptidyl peptidase 4 inhibitors; FDC: Fixed dose combination; FPG: Fasting plasma glucose; GLP-1: Glucagon-like peptide-1; GLUT2: Glucose transporter; HbA1c: Glycated hemoglobin; HF: Heart failure; K+: Potassium ion; RSSDI-ESI: Research Society for the Study of Diabetes in India; SGLT2i: Sodium/glucose cotransporter-2 inhibitors; SU: Sulfonylurea; SUR1: Sulfonylurea receptor 1; TZD: Thiazolidinedione;

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